A Project Report on

ESP32 WEB SERVER-ENABLED IOT SMART FILE LOCKER WITH RFID AUTHENTICATION AND AUTOMATED LOCK CONTROL

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Electronics and Communication Engineering

by

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Acknowledgement

We would like to thank **Dr. Rev. Fr. Joseph C. C.**, Vice Chancellor, CHRIST (Deemed to be University), Bangalore, **Dr. Rev. Fr. Viju P. D.**, Pro Vice Chancellor, **Dr. Fr. Jiby Jose E.**, Director, School of Engineering and Technology, Bangalore, **Rev. Fr. Shijin P. J.**, Assistant Director, School of Engineering and Technology, CHRIST (Deemed to be University), **Dr. Raghunandan Kumar R.**, Dean, and **Dr. Mary Anitha E. A.**, Associate Dean, for their kind patronage.

We would also like to express our sincere gratitude and appreciation to **Dr. Inbanila K**, Head of the Department, Department of Electronics and Communication Engineering, for giving us this opportunity to take up this project.

We express our deepest gratitude to our guide, **Dr. Jesuwanth Sugesh R G**, Assistant Professor, Department of Electronics and Communication Engineering, for his constant support, guidance, and encouragement throughout the project.

We would like to extend our sincere thanks to our panel members, **Dr. Praghash K**, **Dr. Gokuraju Thriveni** and **Dr. Senthilnathan S**, for their valuable feedback, suggestions, and encouragement during the project evaluation.

We are also thankful to our friends and peers for their valuable suggestions and moral support.

Declaration

We, hereby declare that the project titled "**ESP32 WEB SERVER-ENABLED IOT SMART FILE LOCKER WITH RFID AUTHENTICATION AND AUTOMATED LOCK CONTROL**"is a record of original project work undertaken for the award of the degree of **Bachelor of Technology** in **Electronics and Communication Engineering**. We have completed this study under the supervision of **Dr. Jesuwanth Sugesh R G**, Department of Electronics and Communication Engineering.

We also declare that this project report has not been submitted for the award of any degree, diploma, associateship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

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Abstract

The NBA File Locker System is an IoT-based secure document storage system that utilizes RFID technology for authentication and access control using a Radio Frequency Identification (RFID) system for the NBA File Locker System. The system also includes a NodeMCU (ESP8266-based development board) and an Arduino Nano which will allow the system to authenticate users, operate the lockers, and monitor the lockers in real time. The system has two RFID readers; the first reader is used to authenticate a user, and the second reader is used to verify that files have been placed in the locker and are still present. The locker locking and unlocking mechanism utilizes a servo motor that is controlled based on the user's authentication. The system utilizes a Wi-Fi connection and provides access to a web-based dashboard that displays current locker status, current file presence, and user access history in real-time. The system has also implemented a Telegram notification for every time access is granted. The NBA File Locker System is an affordable, secure, and reliable document management solution developed using IoT technology.

***Keywords*:** NBA FIle Locker System; RFID; Node MCU ESP8266; Arduino Nano; Smart Locker; Web Dashboard; IoT security; Servo Motor Locking System; Telegram Notifications.

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GLOSSARY

Item	Description
IoT	**I**nternet **o**f **T**hings – Technology that connects devices to the internet for monitoring and control.
ESP8266	A Wi-Fi enabled microcontroller (NodeMCU) used to control and monitor the smart locker system.
RFID	**R**adio **F**requency **ID**entification – Technology used for user authentication and file detection.
NodeMCU	A development board based on ESP8266 used for IoT applications.
Servo Motor	A motor used to automatically open and close the locker door.
LED	**L**ight **E**mitting **D**iode – Used for visual status indication.
Telegram Bot API	A messaging service used to send automatic reminder notifications after 7 days.
Dashboard	A web-based interface used to monitor locker status, file presence, and user access logs.

Chapter 1

INTRODUCTION

The NBA File Locker System was created to be a high-tech security solution that allows for the safe storage of vital certificates or documents while also providing real-time monitoring of those items. Most storage systems used by businesses for documents do not have secure, real-time monitoring capabilities or automated access control capabilities. This project combines RFID technology with microcontroller technology to create a secure authentication system and retrieve access control to the locker.

To accomplish this, a Node MCU ESP8266 and an Arduino Nano are used for controlling the system, while dual RFID readers are used for each user's authentication and for detecting the files inside the locker. A servo motor is used for automatically locking and unlocking the locker door.

The project also includes a web dashboard to monitor the status of the locker, whether or not any files are currently stored in it, and to display access history of the locker, all of which can be accessed via Wi-Fi. The project will also provide users with access notification updates via the Telegram application.

This project will enable businesses to implement a reliable and fast document management system by combining both hardware and software components at a reasonable cost.

1.1 Problem Formulation

Educational institutions and organizations require secure storage for certificates and other important documents, which are becoming more common as the demand for securely storing such certificates continues to grow. Traditional locker systems do not provide reliable means of authenticating users before they access a locker, automatically locking a locker after a user has accessed it (which would otherwise provide the opportunity for someone to access a locker that should not be accessible), and as a result of monitoring the locker for events happening (such as file placement and/or removal), and also have no way of tracking who accessed their lockers.

Based on these challenges, we have come up with a plan to create a smart locker system that utilizes RFID technology to address these issues. Specifically, this system will ensure that only approved users can access the locker, automatically locks the locker after a user has accessed it, detect when a user places or removes files from the locker, and provide a means of monitoring the locker in real-time through the use of a web interface. We will utilize Node MCU ESP8266, Arduino Nano, RFID reader, and servo motor to develop this efficient and secure system.

1.2 Problem Identification

The problems that have been identified with the current locker systems are as follows:

- No mechanism for user authentication
- No mechanism for automatic locking
- No ability to detect the placement or removal of files from a locker
- No real-time monitoring system
- No record of history access to the locker

These deficiencies demonstrate a critical need for an automated, secure, and monitored locker. Based upon these observations, the problem statement is clear, to design and create an intelligent locker system with secure access, automatic control and real-time monitoring of documents.

1.3 Problem Statement & Objectives

Problem Statement

To create an NBA Files Locking System that protects valuable documents and certificates using an RFID-based authentication system, a system to lock and unlock automatically, and a web-based dashboard for real time display of locker status.

Objectives

- To design a smart locker system for secure storage of certificates and important documents.
- To design a smart locker system for secure storage of certificates and important documents.
- To implement an automatic locking mechanism using a servo motor controlled through RFID-based authorization.
- To create a web dashboard for monitoring locker status, file presence, and user access records.
- To test and validate the system performance in a real-time environment to ensure reliability and security.

1.4 Limitations

Because of the time and resources available for the development of this project, there are certain limitations. Currently, the system supports only a limited number of users since it is developed as a prototype version of the hardware setup. Therefore, these limitations must be addressed before implementing the system on a larger scale or in real-time environments. Another limitation is that all updates to the web dashboard and Telegram notifications depend entirely on a stable Wi-Fi connection. Additionally, both the locker system and the monitoring device must be connected to the same Wi-Fi network for proper communication, which restricts remote access from different networks. Finally, the current version of the project does not include advanced security features such as biometric authentication or cloud-based data storage.

Chapter 2

RESEARCH METHODOLOGY

The research methodology used in developing the NBA File Locker System utilized a structured approach to implement the project. This phase included total hardware and software integration. The methodology included a systematic process to move from background studies and system design through to hardware implementation, dashboard development, and performance evaluation of the overall system. The focus of the project was to develop an automated, secure, and real-time monitored locker system that used RFID authentication and IoT-based communications.

2.1 Literature and Technology Review

A preliminary investigation was conducted into the various types of document storage and locker systems to better assess their current gaps in functionality and to identify areas for improvement. This investigation examined technologies associated with secure authentication, embedded systems, and IoT-based monitoring solutions. The following was a list of technological types that were reviewed:

- Use of RFID-based Authentication (RFIDs) for securely verifying the identity of the end-user.
- Use of a Dual RFID Reader to allow for both user verification and to identify the presence of documents.
- Use of a Servo Motor-based automated Lock and Unlock system.

- Use of a Node MCU ESP8266 for communicating via Wi-Fi and for monitoring lockers in real-time.
- Use of a web-based dashboard for tracking the status and access to each locker.
- Utilization of telegram to automatically notify users with access updates.

Results from this study were used to help formally define the system architecture as well as to identify appropriate hardware and software components required for implementation.

2.2 System Design and Implementation

The system architecture designed based on the identified user requirements included the integration of hardware software components. The implementations carried out as part of this design included:

- Connecting an Arduino Nano to an RFID reader to verify user authentication.
- Programming the Node MCU ESP8266 for wireless communications to send/receive data in real-time.
- Developing the control logic needed to control a servo motor that would unlock-/lock the locker automatically.
- Connecting LED lights to indicate if user access was approved or denied.
- Setting up communication between the hardware components and the web-based user dashboard.

The first RFID Reader was programmed to authenticate an authorized user, and the second RFID Reader was used to determine the presence of an Item(s) inside the locker. The Node MCU provided real-time communications between the locker system and the Monitoring Dashboard.

2.3 Functional Testing and Analysis

After completing implementation of the system's design-plus-development phases, functional testing was completed under different Operational Scenarios to confirm System Performance. The tests performed included:

- Attempts to access the system using authorized credentials.
- Attempts to access the system using unauthorized credentials.
- Detection of files added to and removed from the system.
- Verification of the operation of real-time updates on the dashboard.
- Testing of Telegram notifications sent by the system.

System performance was evaluated based on the following criteria:

- System's response time to authentication requests.
- Accuracy of the detection of files present within the system.
- Reliability of the dashboard data updated by the system.
- Proper processing of lock/unlock requests.

Results of the tests indicated that the NBA File Locker System provides secure authentication,automated locking and real-time monitoring functionality.

2.4 Conclusion

The methodology utilized to create an Integrated Hardware-based NBA Locker System enabled the systematic development of a functional hardware-based NBA Locker System. This now completes the second phase of the development effort, bringing together the RFID authentication, microcontroller, automated locking and IoT Monitoring implementations into an actual working prototype (unlike previous backend simulation efforts). As a complete operation of Hardware, Software and IOT, this system provides the foundation for a reliable, secure, and real-time locker management solution that can be expanded and enhanced in future implementations.

Chapter 3

LITERATURE SURVEY AND REVIEW

Any research-based project requires a detailed literature survey to understand existing systems, technologies, and research gaps. For the NBA File Locker System, relevant research papers and conference publications related to smart lockers, RFID-based access control, IoT-enabled monitoring, and secure storage systems were studied. The collected literature helped in understanding current technologies and identifying improvements that could be implemented in the proposed system.

This chapter is divided into two sub-sections:

- Literature Collection & Segregation
- Critical Review of Literature

3.1 Literature Collection & Segregation

The literature was collected from journals, conference proceedings, and technical publications related to RFID technology, IoT-based locker systems, and smart security solutions. The selected works were categorized based on their application area and technology usage.

1. **IoT-Based Smart Locker Systems:** C. Bharatiraja et al. (2023) presented an IoT-based centralized smart locker using RFID technology. This work focuses on

secure locker management through network connectivity and centralized monitoring.

2. **Smart E-Locker Using IoT:** K. R. Umadi et al. (2023) proposed a smart e-locker system integrating IoT for improved access control and remote monitoring.

3. **Locker System with RFID and Telegram Notification:** M. Fazero et al. (2024) developed a library locker security system integrating RFID, dual camera monitoring, and Telegram notifications for enhanced security.

4. **ESP32-CAM Smart Locker with Cloud Logging:** A. Nair and S. Ramesh (2024) introduced a smart locker using ESP32-CAM with cloud-based logging for improved data storage and remote access.

5. **Smart Locker 2.0 Using IoT and Machine Learning:** S. Chandrappa et al. (2025) proposed an advanced smart locker system integrating IoT and machine learning for enhanced user-friendly public storage.

These research works were studied to understand authentication mechanisms, monitoring systems, hardware integration, and notification techniques.

3.2 Critical Review of Literature

From the collected literature, it is observed that most smart locker systems focus on RFID-based authentication and IoT connectivity for monitoring. Several systems provide centralized access control and remote logging capabilities. Some advanced systems include camera monitoring and cloud-based storage for better security.

However, certain gaps were identified:

- Many systems focus only on user authentication and do not separately detect file placement or removal.
- Some systems depend heavily on cloud infrastructure, increasing system complexity and cost.
- Limited focus is given to low-cost prototype models suitable for academic or institutional implementation.

- Real-time dashboard visualization is not clearly emphasized in all systems.

Based on this review, the NBA File Locker System was designed to address these gaps by:

- Using dual RFID readers for both user authentication and file detection.
- Integrating a web dashboard for real-time monitoring.
- Implementing Telegram notifications for access updates.
- Developing a low-cost and efficient prototype using Node MCU ESP8266 and Arduino Nano.

Thus, the literature survey helped in identifying existing solutions, understanding their strengths and limitations, and guiding the development of a secure, automated, and monitored file locker system.

Chapter 4

ACTUAL WORK

After identifying and formulating the research problem and completing the literature survey, the actual development of the NBA File Locker System was carried out. This chapter describes the methodology followed, hardware and software implementation, system design, prototype development, and testing performed during the project.

4.1 Methodology for the Study

Upon discovering the issue to be researched and completing the literature revision, a practical implementation of the NBA File Locker System took place. This chapter provides an overview of how the system was implemented, as well as details about the methodology used to execute the project; hardware and software integration within the system; overall system design; prototype development; and testing process. The goal of this phase was to create a final product smart locker system including an RFID-based authentication mechanism, automated locking mechanism, ability to monitor the locker in real-time, and a notification mechanism.

4.2 Experimental Work Completed in the Project

This project was carried out in a structured manner using an orderly process for each of the components needed to develop a solution - hardware, software, integration, and validation were all accomplished following an established methodology.

The project began with identifying research gaps where the needs of the system were clearly defined. All of those needs created system requirements based on both the identified gaps in technology such as real-time monitoring, structured event logging, and secure authentication.

The hardware architecture used a Node MCU ESP8266, Arduino Nano with two RFID readers, a servo motor and two LED indicators and was designed in four distinct phases:

- Interfacing Arduino Nano and two RFID readers to provide the ability to perform user authentication and identify the location of the file/files.
- Writing programming code to control the servo motor to lock/unlock automatically.
- Configuring the Node MCU ESP8266 for WiFi connectivity.
- Establishing communication between all hardware components and sending the data collected to the Web Dashboard.
- Integrating with Telegram as the method of sending access notifications.

By following this methodology, it was ensured that the functions of authentication, control and monitoring were all integrated in a logical and systematic manner.

4.3 Modeling, Analysis & Design

The system was designed using a modular architecture approach. The major modules include:

Module for Authentication - the Authentication process (of users) is accomplished by means of RFID (Radio Frequency Identification).

Module for File Detection - this module will detect whether a file has been inserted into or removed from the locker.

Module for Control - this module is responsible for controlling the Locking and unlocking Servo Motors.

Module for Communication - this module makes use of the Node MCU ESP8266 to communicate via Wi-Fi to the Cloud to send and receive data.

Module for Monitoring - this module has a web dashboard to provide status on the locker as well as to provide access and timestamp records to the user.

The block diagram based approach was used in the design process to portray the interface between the hardware and software components and how they will work together. The Arduino Nano processes RFID inputs and controls the servo motor, while the Node MCU transmits locker status and access logs to the dashboard. The NODE MCU will communicate either over Wi-Fi to the cloud for storing and transmitting the locker status and access logs. LEDs will provide visual feedback to the user of access granted or denied. The Modular Architecture based design will contribute to improving the reliability of the System and add to its Scalability and Maintainability.

4.4 Prototype & testing

A working prototype of the NBA File Locker System was developed using the selected hardware components. The hardware components were assembled and interfaced according to the system design.

The prototype supports:

- RFID-based user authentication
- Automated servo-based locking mechanism
- File presence detection
- Real-time dashboard monitoring
- Telegram-based notification alerts

Testing was performed under various scenarios:

1. Authorized access attempt
2. Unauthorized access attempt
3. File insertion detection
4. File removal detection

5. Network connectivity validation

The prototype demonstrated stable performance, accurate authentication, and reliable real-time monitoring. The system successfully integrates hardware-level implementation with IoT-based monitoring, addressing the gaps identified in the literature review.

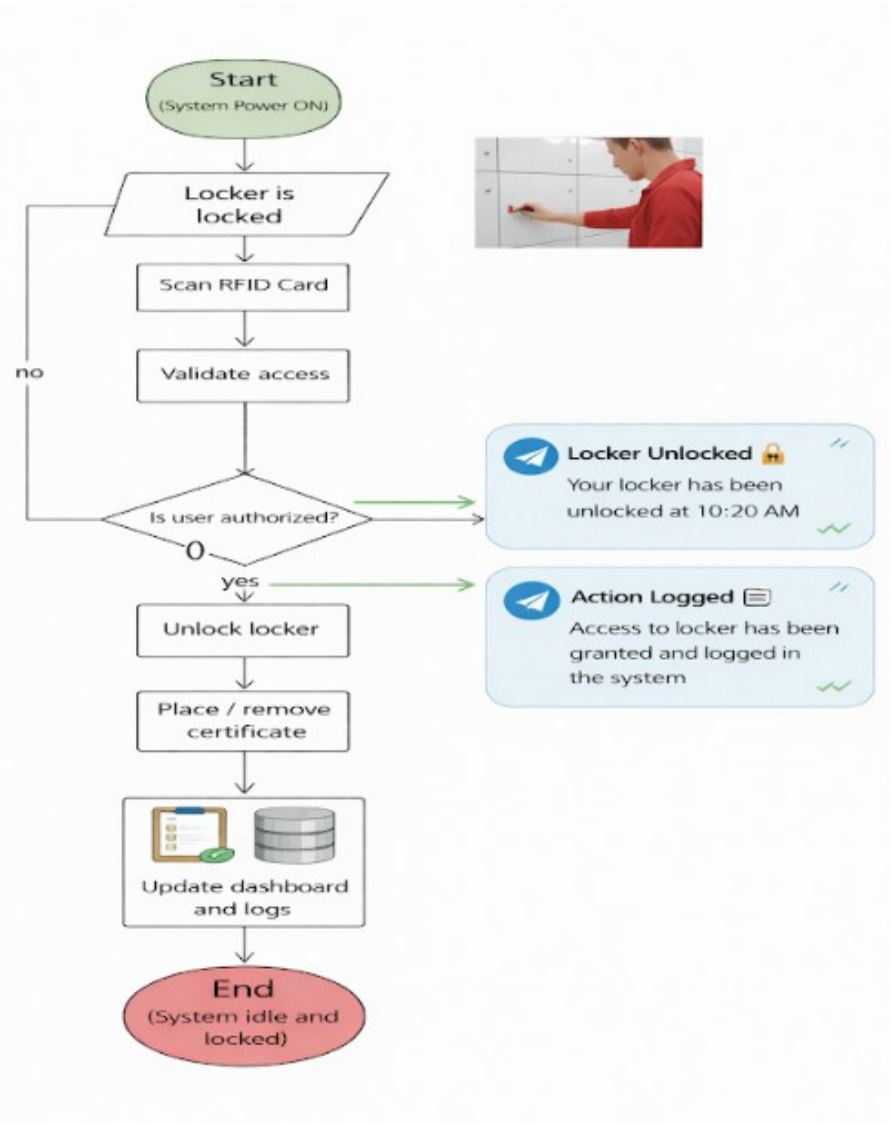

FIGURE 4.1: Flowchart of NBA File Locker System Working Process

The reasons listed below provide a step-by-step illustration of how the NBA File Locker System works, as shown in Figure 4.1:

- The locker will be locked by default as soon as the player first turns on the system.

- When a card holder scans an RFID card at the scanner, the system will determine if they are authorized to access the locker or not.
- If an authorized user is identified, then the servo motor will unlock.
- The user can now put their certificate in and/or take their certificate out of the locker (and/or retrieve any stored documents from the locker).
- Once the user has completed their operation, the NBA File Locker system will update its inventory dashboard/logs through the server.
- Once the update is complete, the locker will return to idle mode and lock itself again.
- If the user is not authorized, access will be denied and the locker will remain locked.

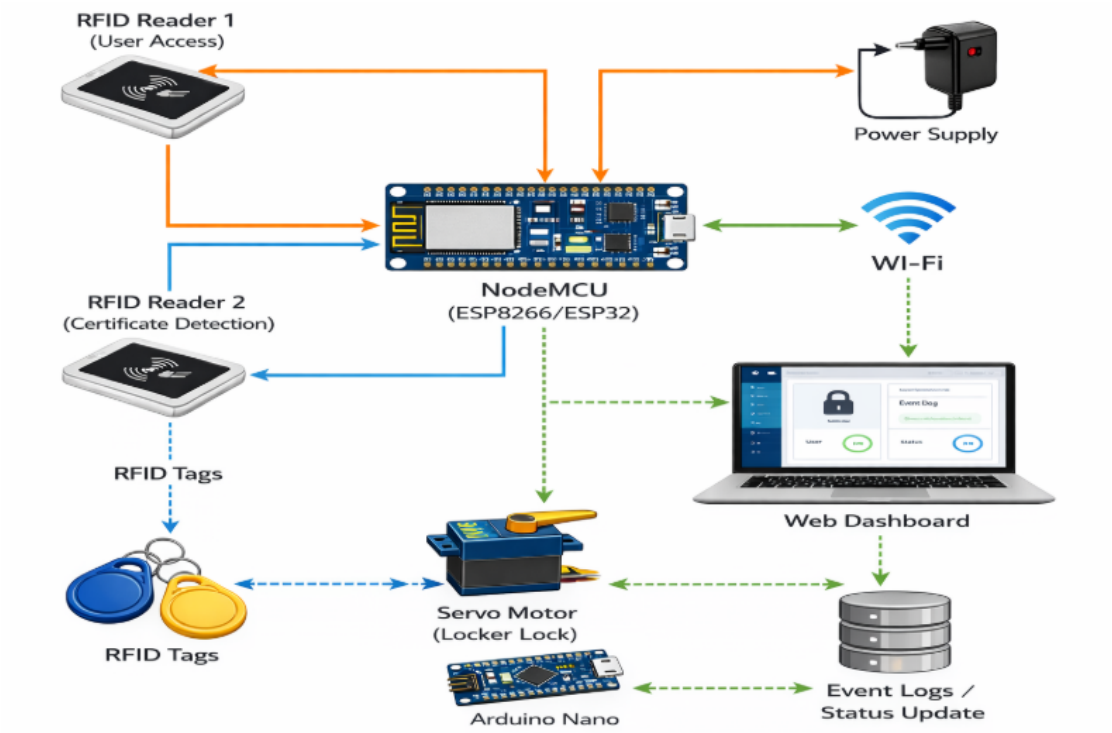

FIGURE 4.2: Hardware–Software Architecture of the NBA File Locker System

As shown in Figure 4.2, the diagram represents the overall hardware architecture and working of the NBA File Locker System. The NodeMCU acts as the main controller of the system. RFID Reader 1 is used for user authentication, while RFID Reader 2 is

used for certificate detection inside the locker. When a valid RFID tag is scanned, the NodeMCU processes the data and controls the servo motor to lock or unlock the locker accordingly.

The system is powered by a regulated power supply, and Wi-Fi connectivity enables communication with the web dashboard. Through Wi-Fi, locker status, file presence, and event logs are updated in real time. The Arduino Nano assists in controlling the locking mechanism and handling related operations. Overall, Figure 4.2 illustrates how authentication, automatic locking, and IoT-based monitoring are integrated to create a secure smart locker system.

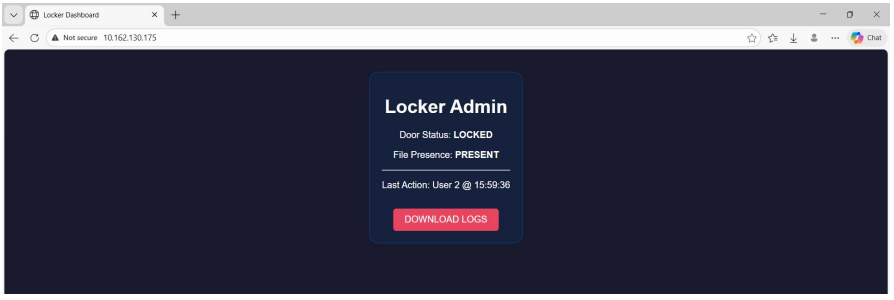

FIGURE 4.3: Web Dashboard Interface Showing Real-Time Locker Status and Access Logs

As shown in Figure 4.3, the illustrative representation displays the web-based administrator dashboard of the NBA File Locker System. It shows the current status of each locker door (Locked), the file presence inside the locker (Present), and the timestamp of the most recent user action. The dashboard enables real-time monitoring of locker activity through Wi-Fi connectivity. Additionally, as illustrated in Figure 4.3, the "Download Logs" feature allows administrators to retrieve previously recorded locker usage data for tracking, monitoring, and security purposes.

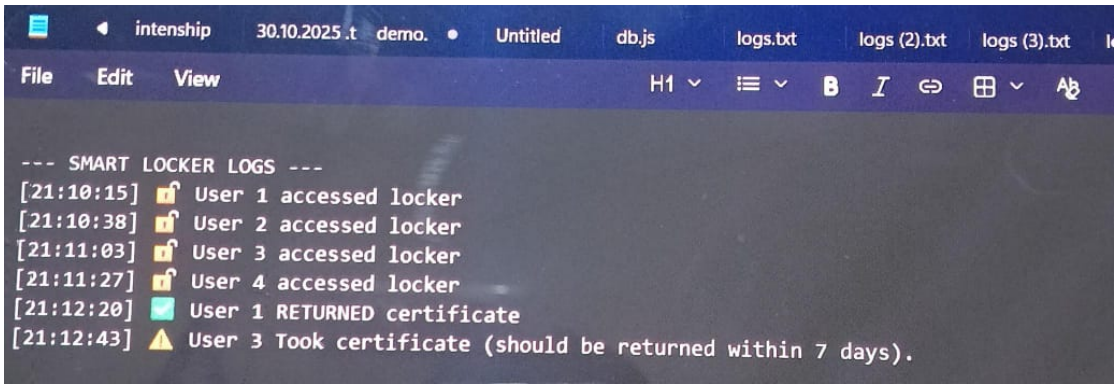

FIGURE 4.4: Smart Locker System Log File Showing User Access and Certificate Activity

As shown in the Figure 4.4, the image displays the text-based log records of the Smart Locker System. Each entry includes the time stamp and details of user actions such as accessing the locker, returning a certificate, or taking a certificate. The logs help in tracking user activity, maintaining security, and monitoring certificate handling within the system.

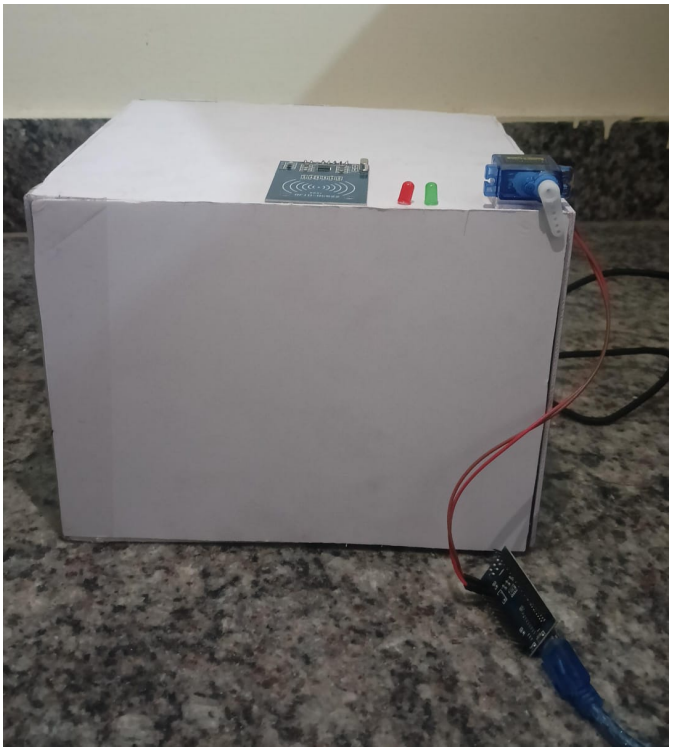

FIGURE 4.5: Prototype Model of the NBA File Locker System with RFID Module and Servo Lock Mechanism

As shown in Figure 4.5, the image represents the external working model of the NBA File Locker System. An RFID card reader is positioned on the top of the locker to scan

user cards for authentication before granting access. The green and red LED indicators display whether access is approved or denied. The locking and unlocking mechanism of the locker is controlled by a servo motor mounted on the front side of the system. Overall, Figure 4.4 demonstrates the practical prototype implementation of the system for secure locker access.

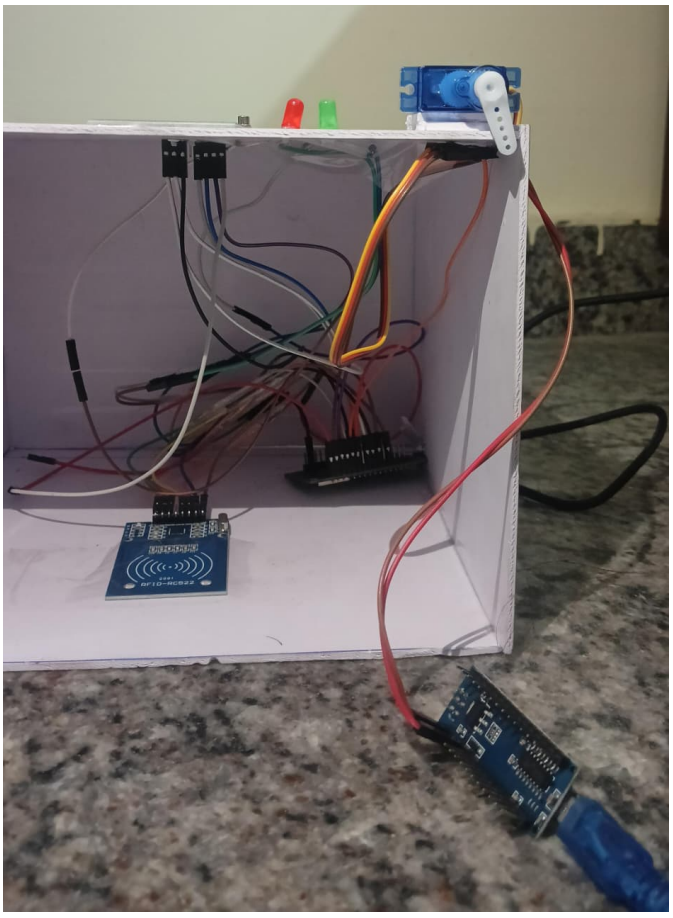

FIGURE 4.6: Internal Wiring and Component Setup of the NBA File Locker System

As shown in Figure 4.6, the image represents the internal working of the NBA File Locker System. The visible components include the Arduino Nano, NodeMCU, RFID module, servo motor connections, and jumper wires that interconnect all parts of the system. These components work together to authenticate users, control the locking mechanism, and enable data communication. Figure 4.5 illustrates how the internal circuitry ensures secure access control and continuous monitoring of locker activities.

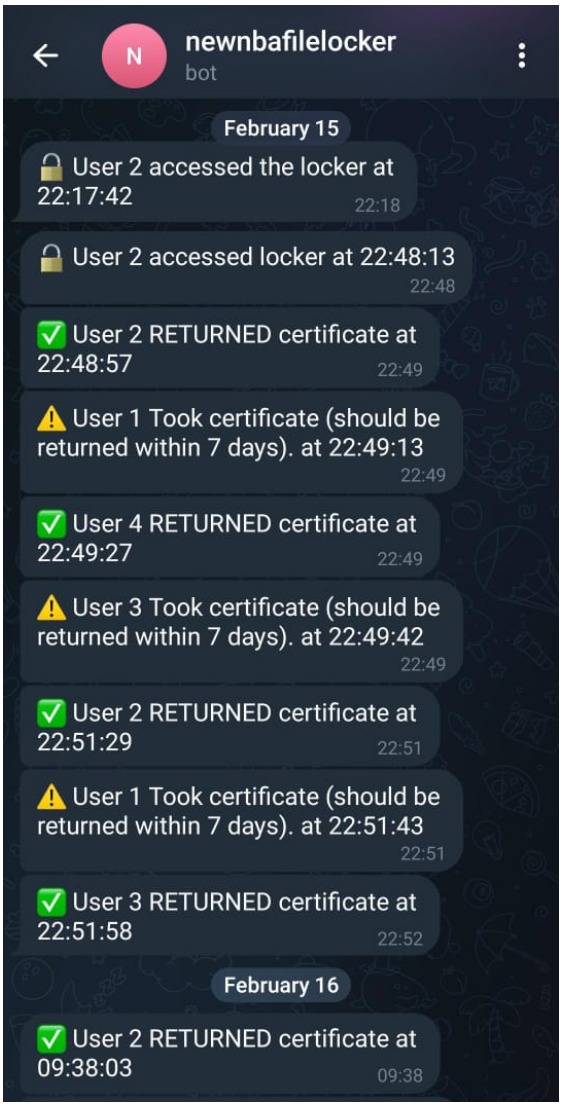

FIGURE 4.7: Telegram Bot Message Showing Locker Access Update

As shown in Figure 4.7, the image displays Telegram notifications generated by the NBA File Locker System. The notifications include user access details along with the exact time the action was performed. These real-time alerts enable monitoring of locker usage and provide immediate updates whenever the locker is accessed or files are handled, ensuring better tracking and security.

FIGURE 4.8: Telegram Bot Message Showing Locker Access Update

As shown in Figure 4.8, the image displays Telegram notifications generated by the NBA File Locker System. The notifications include user access details along with the exact time the action was performed. These real-time alerts enable monitoring of locker usage and provide immediate updates whenever the locker is accessed or files are handled, ensuring better tracking and security.

Chapter 5

RESULTS, DISCUSSIONS AND CONCLUSIONS

This chapter presents the results obtained from the development and testing of the NBA File Locker System. The performance of the system is analyzed based on security, reliability, and functionality. The outcomes are discussed in relation to the objectives defined earlier. A comparative study with existing systems from the literature review is also included. Finally, conclusions and scope for future work are presented.

5.1 Results & Analysis

The NBA File Locker system has now been properly tested and operated under many different access condition scenarios. During the testing phase, the performance of the system and its ability to authenticate, monitor, and run automatically worked as expected.

- Regarding RFID authentication, authorized users were able to gain access to the locker, while unauthorized users were prevented from doing so.
- The dual RFID reader configuration for identifying and determining whether or not there was a file present worked appropriately to confirm each user's identity and the file.

- The system was able to successfully process operations using two working systems, Node MCU ESP8266 and Arduino Nano, and successfully operated a locking/unlocking process using a servo motor with each operation providing a level of security for documents.
- Additionally, there were green and red LED indicators that provided immediate locations and visuals for users regarding their access status.
- The Web Dashboard provided real-time reporting regarding the status of the locker, whether or not there was a file present, and the access logs for each locker/operator.
- The use of Telegram notifications successfully recorded and reported to users the access to any locker or activity related to any file.

Overall, the repeated testing confirmed that the system's operation was stable and consistent and validated the reliability of the proposed solution.

5.2 Comparative Study

When comparing our proposed project with those in literature review, clearly show that our system, an NBA File Locker System, enhances all existing systems based on the following 3 criteria:

1. Traditional lock-up systems utilize single RFID readers for both user authentication and file detection; our proposal utilizes dual RFID readers which separate user authentication and file detection, which enhances the accuracy of both processes.
2. The inclusion of a live web-based dashboard will provide real-time monitoring capabilities compared to systems that only utilize local displays.
3. As opposed to most traditional systems, our project provides access to remote monitoring capabilities via telegram notifications.

Although some state-of-the-art systems have features such as camera support and cloud computation; our proposal provides a system that utilizes only essential security and monitoring capabilities while remaining within strict cost constraints for educational use.

5.3 Discussions

Using an RFID-based approach to control access to secure repository for storing documents meets the requirement of secure storage. Hardware components including RFID cards, microcontrollers, servos and LEDs worked well as one system.

The web dashboard provides real-time visibility into system status and also provides better control over the system. A locking mechanism was built to ensure that each time someone accesses a locker, it will remain locked at all times. While the number of users currently supported is limited, accessing the dashboard update and receive notifications are dependent on stable Wi-Fi connection. Both these factors demonstrate potential areas for future improvement.

5.4 Conclusions

The NBA's design and installation of the File Locker System has been a success and now provides a solution for securely storing certificates and other important documents. All objectives have been met as part of this project, including; secure entry via RFID-based authentication, automatic locking through a servo, real-time monitoring via a web dashboard and notifications via Telegram.

Overall, the performance of the system during the testing phase was consistent and demonstrated high reliability, very low cost, and an efficient system. These results support the conclusion that the proposed idea is a viable option for academic and institutional applications, in which the secure management of documents is required.

5.5 Scope for Future Work

The NBA File Locker System can be further improved and enhanced in several ways:

- Increase the number of users who can access the locker.
- Improve system speed and overall performance.
- Enhance the dashboard interface for better clarity and usability.
- Conduct testing with more users and multiple trials.
- Improve the locker opening and closing mechanism for smoother operation.
- Integrate advanced security features such as biometric authentication.
- Implement cloud-based data storage for better scalability.

With these improvements, the system can be developed into a more advanced and scalable smart locker solution suitable for large-scale institutional or organizational deployment.

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Appendix A

Source Code of NBA File Locker System

This appendix contains the complete Arduino program used for implementing the NBA File Locker System using Node MCU ESP8266, dual RFID readers, servo motor, web dashboard, and Telegram notification integration.

A.1 Complete Arduino Program

```
#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <ESP8266WebServer.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include <NTPClient.h>
#include <WiFiUdp.h>

// ---------------- NETWORK ----------------
const char* ssid =  vivo Y21 ;
const char* password =  7418922390 ;
const char* BOTtoken =  8482210850:AAGg0cJwqTNVilV6hHbN0-↩
   J3XhToX0yDqwU ;
```

```
const char* CHAT_ID =  996053949 ;

// ---------------- USERS ----------------
const String USER_1_UID =  85 A1 39 06 ;
const String USER_2_UID =  9E D7 F3 05 ;
const String USER_3_UID =  A3 DF 37 27 ;
const String USER_4_UID =  63 11 C0 27 ;

// ---------------- CERTIFICATES ----------------
const String CERT_UID   =  C0 D5 9E 5F ;
const String CERT_UID_1 =  03 B5 9E E4 ;
const String CERT_UID_2 =  33 4F B4 E4 ;

// ---------------- PINS ----------------
#define SS_1_PIN D0
#define SS_2_PIN D1
#define RST_PIN  D3
#define SERVO_PIN D4
#define LED_G     D2
#define LED_R     3
#define BUZZER    D8

MFRC522 rfid_door(SS_1_PIN, RST_PIN);
MFRC522 rfid_file(SS_2_PIN, RST_PIN);
Servo lockerServo;
ESP8266WebServer server(80);

WiFiClientSecure client;
UniversalTelegramBot bot(BOTtoken, client);
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP,  pool.ntp.org , 19800);

// ---------------- STATE ----------------
String lockerStatus =  LOCKED ;
String fileStatus =  Checking... ;
String lastUser =  None ;
String lastActionTime =  N/A ;
String eventLogs =  --- SMART LOCKER LOGS ---\n ;

// ------------------------------------------------
// TELEGRAM SEND
// ------------------------------------------------
void sendTelegram(String msg) {
  server.stop();
```

```
  digitalWrite(SS_1_PIN, HIGH);
  digitalWrite(SS_2_PIN, HIGH);

  client.setInsecure();
  client.setBufferSizes(512, 512);

  if (bot.sendMessage(CHAT_ID, msg,   )) {
    Serial.println( Telegram Sent );
  } else {
    Serial.println( Telegram Failed );
  }

  client.stop();
  server.begin();
}

// ----------------------------------------------
// WEB DASHBOARD
// ----------------------------------------------
void handleDownload() {
  server.sendHeader( Content-Disposition ,  attachment; filename↩
   =logs.txt );
  server.send(200,  text/plain , eventLogs);
}

void handleRoot() {
  String html =  <html><head><meta name='viewport' content='↩
   width=device-width, initial-scale=1'><style> ;
  html +=  body { font-family: sans-serif; text-align: center; ↩
   background: #1a1a2e; color: white; } ;
  html +=  .card { background: #16213e; padding: 20px; border-↩
   radius: 15px; display: inline-block; margin-top: 30px; border↩
   : 2px solid #0f3460; } ;
  html +=  .btn { background: #e94560; color: white; padding: 10↩
   px 20px; text-decoration: none; border-radius: 5px; display: ↩
   inline-block; margin-top: 15px; } ;
  html +=  </style><title>Locker Dashboard</title></head><body><↩
   div class='card'><h1>Locker Admin</h1> ;
  html +=  <p>Door Status: <b>  + lockerStatus +  </b></p> ;
  html +=  <p>File Presence: <b>  + fileStatus +  </b></p> ;
  html +=  <hr><p>Last Action:   + lastUser +   @   + ↩
   lastActionTime +  </p> ;
  html +=  <a href='/download' class='btn'>DOWNLOAD LOGS</a></↩
   div></body></html> ;
```

```
  server.send(200,  text/html , html);
}

// ------------------------------------------------
// SETUP
// ------------------------------------------------
void setup() {
  Serial.begin(115200);

  pinMode(LED_G, OUTPUT);
  pinMode(LED_R, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  pinMode(SS_1_PIN, OUTPUT);
  pinMode(SS_2_PIN, OUTPUT);
  digitalWrite(SS_1_PIN, HIGH);
  digitalWrite(SS_2_PIN, HIGH);

  lockerServo.attach(SERVO_PIN);
  lockerServo.write(0);

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print( . );
  }

  server.on( / , handleRoot);
  server.on( /download , handleDownload);
  server.begin();

  timeClient.begin();

  SPI.begin();
  rfid_door.PCD_Init();
  rfid_file.PCD_Init();

  Serial.println( \nConnected. IP: );
  Serial.println(WiFi.localIP());
}

void loop() {
  server.handleClient();
}
```